

Nandan Varma Santhosh

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EDUCATION

University of Waterloo

Sep 2024 – Apr 2029

Honours BAsc. Mechatronics Engineering | Artificial Intelligence Minor - **3.86 GPA**

Waterloo, ON

- **Relevant Coursework:** Digital Logic, Circuits, Data Structures & Algorithms, Sensors & Instrumentation

TECHNICAL SKILLS

Embedded Systems: STM32, ESP32, C2000, Arduino, RTOS, SPI, USART, CAN, PWM, PID, ROS2, I²C

Languages: Python, C/C++, VHDL, FPGA, Ladder Logic

Electrical Design/Validation: Altium, KiCAD, LTspice, Oscilloscope, Logic Analyzer, MATLAB

Software: OpenCV, PyTorch, YOLOv5, React, Pandas, NumPy, Git, Linux, VS Code, STM32CubeIDE

EXPERIENCE

Mechatronics Engineer Intern

Jan 2026 – Apr 2026

Acceleration Consortium

Toronto, ON

- Designed a **PLC-based** closed-loop **PID** control system on a PLCNext controller using **ladder logic** and **Structured Text**, achieving ± 0.3 °C accuracy, **25** min settling time, and **>95%** steady-state stability in a thermal process.
- Developed **STM32** firmware in **C** on **CubeIDE** using **SPI-based** motor driver interfaces, **PWM**, and implementing **interrupt-driven** command prioritization for **motion control** in an automated gantry system.
- Engineered **PID** temperature control firmware in **C** with **FreeRTOS**, implementing a **10Hz** control loop with **5-sample** averaging filter and real-time **Modbus TCP** data streaming for remote process monitoring.
- Fabricated a multi-axis motor driver **PCB** in **Altium Designer**, managing **4+** layers, **100+** components, and full **DFM** review through **JLCPCB** fabrication, enabling motor driver integration for the gantry system.

Embedded Systems Research Assistant

Sep 2025 – Dec 2025

University of Waterloo Department of Mechanical and Mechatronics Engineering

Waterloo, ON

- Built a **Raspberry Pi** kiln controller integrating a **thermocouple**, **SSR**, and isolated high-voltage stage, achieving ± 2 °C precision at **240 VAC**.
- Developed a **Python PID control** system with a web dashboard for real-time monitoring and firing schedules, reducing overheating by **40%**.

Software Engineer Intern

May 2025 – Aug. 2025

Quasar Consulting Group

Toronto, ON

- Deployed an **Azure OpenAI** chatbot with **C#**, **.NET** and **Autodesk Revit** integration, leveraging **semantic search** to cut support tickets by **35%**, halved lookup time, and accelerate onboarding for **20+** engineers.
- Developed a Microsoft Word Add-in using **Node.js**, **npm**, **Office JavaScript API**, and **TypeScript** with **5+** automated features, reducing manual formatting by **40%** for **200+** users across 8 sectors.
- Built **10+** cloud functions in **Python** and **TypeScript (Deno)** with **PostgreSQL** and **Firebase**, automating repetitive workflows and improving data delivery performance by **50%**.

PROJECTS

3-Phase BLDC Motor Inverter PCB | Altium Designer, STM32, MOSFETs

Oct 2025

- Designed and implemented the three-phase inverter on **Altium Designer** using 6 N-channel **MOSFETs** with half-bridge **gate drivers** for efficient **BLDC motor** commutation.
- Developed **STM32-based PWM** control firmware generating complementary **20kHz** switching signals with adjustable **dead-time** for six-step commutation, achieving precise motor speed and torque regulation.

InspectPCB | Python, OpenCV, PyTorch, YOLOv5, StreamLit

Jun 2025

- Developed a **PCB defect detection** tool using **OpenCV** and **Python**, enabling identification of **5+ defect types** (e.g., solder bridges, missing components) with **90%** accuracy on test datasets.
- Trained and deployed a **YOLOv5 computer vision** model on **1,200+** PCB images (DeepPCB), achieving **92.3% mAP**, and integrated it into a **StreamLit** web app for interactive PCB inspection.